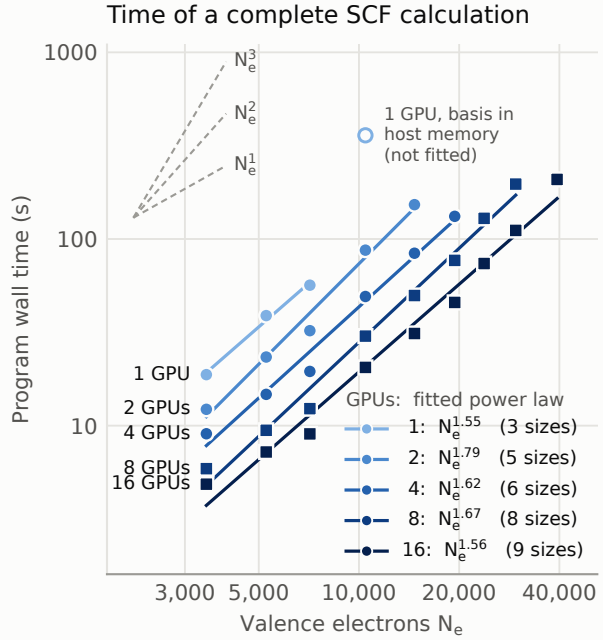
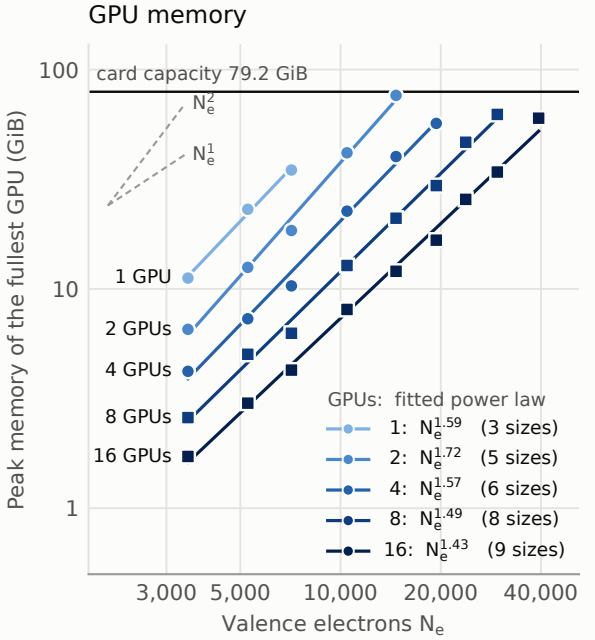

Time of a complete SCF calculation
Program wall time (s)
1000
100
10
N_e^3
N_e^2
N_e^1
1 GPU, basis in host memory (not fitted)
1 GPU
2 GPUs
4 GPUs
8 GPUs
16 GPUs
GPUs: fitted power law
1: $N_e^{1.55}$ (3 sizes)
2: $N_e^{1.79}$ (5 sizes)
4: $N_e^{1.62}$ (6 sizes)
8: $N_e^{1.67}$ (8 sizes)
16: $N_e^{1.56}$ (9 sizes)
3,000
5,000
10,000
20,000
40,000
Valence electrons N_e


GPU memory
Peak memory of the fullest GPU (GiB)
100
10
1
card capacity 79.2 GiB
N_e^2
N_e^1
1 GPU
2 GPUs
4 GPUs
8 GPUs
16 GPUs
GPUs: fitted power law
1: $N_e^{1.59}$ (3 sizes)
2: $N_e^{1.72}$ (5 sizes)
4: $N_e^{1.57}$ (6 sizes)
8: $N_e^{1.49}$ (8 sizes)
16: $N_e^{1.43}$ (9 sizes)
3,000
5,000
10,000
20,000
40,000
Valence electrons N_e